Milène Mandeville

Applied Mathematics Student (She/Her)

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Languages

- French (native)
- Dutch (B1)
- English (C1)
- German (B1)

Programming skills MileneMdvl

- Python (advanced)
- Julia (intermediate)
- LaTex (advanced)
- Matlab (intermediate)

Education

- **MSc Applied Mathematics**, Rijksuniversiteit Groningen (2024–2026)

  Relevant coursework: Computational Fluid Dynamics, Finite Element Methods, Coupled Problems, Inverse Problems in Imaging, Numerical Simulation of Turbulence, Randomised Numerical Linear Algebra

- **BSc Applied Mathematics**, Rijksuniversiteit Groningen (2020-2024)

Research

- **Master Thesis**: University of Groningen (2026, ongoing)
  - **(Working) Title**: Extension of a 1D model for large eddy simulation in Burger's turbulence
  - **Supervisors** prof. dr. ir. R.W.C.P Verstappen and dr. ir. R. Luppes
  - **Description**: The aim of this work is to extend an existing model (Verstappen, 2026) for large eddy simulation in Burger's turbulence. This one dimensional model relies on a uniform mesh with periodic boundary conditions, and we wish to generalise it to a three dimensional triangular grid.
- **Internship**: Von Karman Insitute for Fluid Dynamics (2026)
  - **Title**: Towards Efficient Galerkin Projection in Lagrangian ROMs for Sloshing
  - **Supervisors**: prof. M. A. Mendez, ir. S. Ahizi, dr. J. Koellermeier, and dr. ir. R. Luppes,
  - **Description**: In this paper I derive a reduced order model for smoothed particle hydrodynamics, by rewriting the standard neighbour sums as dense matrix operations, and using proper orthorgonal decomposition and Galerkin projection to obtain a linear time invariant model. However, the kernel transformations inherent to particle dynamics introduce a nonlinear state dependent term. To this end, I evaluate three methods: linearisation, random Fourier features, and a singular value decomposition based interpolation.
- **Bachelor Thesis**: University of Groningen (2023)
  - **Title**: Average Surface Roughness for 2-Dimensional Atmospheric Flow
  - **Supervisors**: dr. ir. R. Luppes and dr. A. E. Sterk
  - **Description**: In this paper I present a study of four different methods of computing an average (effective) roughness length (following André and Blondin (1986), Taylor (1989), de Vries et. al. (2003), and taking the average of the logarithmic law).

Professional experience

- **Acadomia (tutoring company): online teacher** (2024–present)
  - Teaching mathematics to middle school and high school students in French
- **University of Groningen: teaching assistant** (2023–present)
  For the following Bachelor courses, I gave tutorials and/or computer labs in Python. I also graded homeworks and exams, and evaluated students with oral examinations for the courses with labs.
  - Mathematical Tools for BME (2025) – C. Karakurt, PhD
  - Project Mathematical Modelling (2025) – prof. dr. ir. R. Verstappen
  - Computational Methods of Science (2024–2026) – prof. dr. ir. R. Verstappen, dr. ir. R. Luppes
  - Mathematical Modelling (2023–2026) – dr. S. Trenn, prof dr. ir. R. Verstappen
  - Advanced Systems Theory (2023) – prof. M.K. Camlibel

Other experience

- **LaTex Workshop** (2025-2026)
  - Teaching first year Bachelor Students how to use LaTex
  - Focusing on equations, theorems, figures, tables, BibTex management and Beamer
  - Introduction to TikZ
- **Secretary of a committee of the FMF (study association)** (2023–present)
  - Writing minutes summarising the meetings
  - Writing emails inviting professors to give a talk
  - Organising the annual integration competition and a series of informal lectures

References

- Prof. dr. ir. Roel Verstappen, University of Groningen, r.w.c.p.verstappen@rug.nl
- Dr. ir. Roel Luppes, University of Groningen, r.luppes@rug.nl
- Prof. Miguel Mendez, von Karman Institute for Fluid Dynamics, miguel.alfonso.mendez@vki.ac.be

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